

Straight lines - Set 2

TMUA practice questions by topic. Specification MM3.1, M4.9, M4.10. 3 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The lines $y = mx + 4$ and $y = 3x - 2$ intersect at a point with positive x - and y -coordinates. Find the complete set of values of m .
- A** $m < 3$
B $m < -6$ or $m > 3$
C $-6 < m < 3$
D $m > -6$
E $0 < m < 3$
F $-6 < m < 0$
- 2 The distance between the parallel lines $3x + 4y = 5$ and $3x + 4y = 20$ is
- A** 15
B 3
C 4
D $\frac{15}{7}$
E 1
F 5
- 3 The line through $(a, 0)$ and $(0, b)$, with $a, b > 0$, passes through $(2, 3)$. Which of the following statements is/are true?
- I** $\frac{2}{a} + \frac{3}{b} = 1$
II $a > 2$ and $b > 3$
III The smallest possible value of ab is 24
- A** none of them
B I only
C II only
D III only
E I and II only
F I and III only
G II and III only
H I, II and III